

MODULE 4

Vector Integral Calculus

Mathematics-I — MA101

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Syllabus Coverage

Topic
Surface integrals — definition, evaluation and physical applications
Volume integrals — definition and evaluation
Green's theorem in the plane
Gauss Divergence theorem and its physical significance
Stokes' theorem and its physical significance

Chapter 1: Surface Integrals

A surface integral extends the idea of a double integral to a curved surface S in space. It is the natural tool for measuring how a scalar quantity accumulates over a surface, or how a vector field ‘flows through’ a surface — its flux.

1.1 Definition of a Surface Integral

Let S be a piecewise smooth surface and $\mathbf{n}$ the unit outward normal at a point of S . For a vector function $\mathbf{F}(x, y, z)$ continuous on S , the surface integral of $\mathbf{F}$ over S (the flux integral) is defined as:

$$\iint_S \mathbf{F} \cdot \mathbf{n} \, dA$$

If S is given by $z = f(x, y)$ over a region R in the xy -plane, the surface integral can be reduced to a double integral over R by writing dA in terms of $dx \, dy$:

$$\iint_S \mathbf{F} \cdot \mathbf{n} \, dA = \iint_R \mathbf{F} \cdot \mathbf{n} \sqrt{1 + f_x^2 + f_y^2} \, dx \, dy$$

Worked Example — Flux Through a Plane Surface

Evaluate $\iint_S \mathbf{F} \cdot \mathbf{n} \, dA$ where $\mathbf{F} = z \mathbf{i} + x \mathbf{j} - 3y^2z \mathbf{k}$ and S is the surface $x + 2y + 2z = 6$ in the first octant, cut off by the plane $z = 0$.

Solution:

Here $f(x, y) = (6 - x - 2y)/2$, so $f_x = -1/2$, $f_y = -1$. The unit normal to the plane $x + 2y + 2z = 6$ is $n = (1/3)(i + 2j + 2k)$. Then:

$$F \cdot n = (1/3)(z + 2x - 6y^2z)$$

Since $\sqrt{1+f_x^2+f_y^2} \, dx \, dy = 3 \, dz \dots$ — the constant 3 from the normal cancels the Jacobian factor of 3, and the flux reduces to a straightforward double integral over the triangular base of the plane in the xy -plane; carrying this through and integrating over the triangle $0 \leq x \leq 6$, $0 \leq y \leq (6 - x)/2$ gives the value of the flux.

1.2 Physical Applications of Surface Integrals

Application	Meaning
Flux of a vector field	Net quantity of a field (fluid, electric, magnetic) crossing S per unit time
Mass of a curved lamina	$\iint_S \rho(x, y, z) \, dA$, where ρ is surface density
Centre of mass / moments of a shell	Weighted surface integrals of position over S
Electric flux (Gauss's law)	$\iint_S \mathbf{E} \cdot \mathbf{n} \, dA$ equals enclosed charge over ϵ_0

Chapter 2: Volume Integrals

2.1 Definition

Let T be a bounded closed region in space and $f(x, y, z)$ a scalar function continuous on T . Subdividing T into small volume elements ΔV_k and forming the Riemann sum $\sum f(x_k, y_k, z_k) \Delta V_k$, the limit as the subdivision is refined is the volume integral:

$$\iiint_T f(x, y, z) \, dV = \iiint_T f(x, y, z) \, dx \, dy \, dz$$

In cylindrical coordinates (r, θ, z) , $dV = r \, dr \, d\theta \, dz$; in spherical coordinates (r, θ, ϕ) , $dV = r^2 \sin\phi \, dr \, d\theta \, d\phi$. Choosing the coordinate system that matches the symmetry of T greatly simplifies evaluation.

Worked Example — Volume Integral over a Cylinder

Evaluate $\iiint_T x^2 \, dV$ where T is the solid cylinder $x^2 + y^2 \leq a^2$, $0 \leq z \leq b$.

Solution:

Using $x = r \cos\theta$, $dV = r \, dr \, d\theta \, dz$:

$$\iiint x^2 \, dV = \int_{[z=0 \rightarrow b]} \int_{[\theta=0 \rightarrow 2\pi]} \int_{[r=0 \rightarrow a]} r^2 \cos^2\theta \cdot r \, dr \, d\theta \, dz = b \cdot \pi \cdot (a^4/4) = \pi a^4 b/4$$

2.2 Physical Significance

- Total mass of a solid: $m = \iiint_T \rho(x, y, z) \, dV$, where ρ is volume density
- Total charge enclosed in a region, for a given charge density
- Centre of mass and moments of inertia of a solid body

- The right-hand side of the Gauss Divergence theorem, connecting flux to the divergence integrated over the enclosed volume

Chapter 3: Green's Theorem in the Plane

3.1 Statement

Let R be a region in the xy -plane bounded by a piecewise smooth simple closed curve C , traversed so that R lies on the left (positive/counter-clockwise orientation). If $P(x, y)$ and $Q(x, y)$ have continuous first partial derivatives in a domain containing R , then Green's theorem states:

$$\oint_C (P \, dx + Q \, dy) = \iint_R (\partial Q / \partial x - \partial P / \partial y) \, dx \, dy$$

Green's theorem converts a line integral around a closed plane curve into a double integral over the region it encloses, and is the two-dimensional counterpart of both the Divergence theorem and Stokes' theorem.

Worked Example — Verifying Green's Theorem

Evaluate $\oint_C (xy \, dx + x^2 \, dy)$ where C is the boundary of the square $0 \leq x \leq 1$, $0 \leq y \leq 1$, oriented counter-clockwise.

Solution:

Here $P = xy$ and $Q = x^2$, so $\partial Q / \partial x - \partial P / \partial y = 2x - x = x$. By Green's theorem:

$$\oint_C (xy \, dx + x^2 \, dy) = \int_0^1 \int_0^1 x \, dx \, dy = \int_0^1 (1/2) \, dy = 1/2$$

3.2 Applications of Green's Theorem

Application	Formula
Area enclosed by C	$A = (1/2) \oint_C (x \, dy - y \, dx)$
Verifying / replacing a line integral	Convert $\oint_C (P \, dx + Q \, dy)$ into a double integral when that is easier to evaluate
Special case of Stokes' theorem	Treat R as a flat surface in the xy -plane with unit normal $\mathbf{k}$; Green's theorem is Stokes' theorem restricted to a planar field $\mathbf{F} = P \mathbf{i} + Q \mathbf{j}$

Note: Taking $\mathbf{F}(x, y, z) = P(x, y) \mathbf{i} + Q(x, y) \mathbf{j} + 0\mathbf{k}$ and D a flat region in the xy -plane oriented by $\mathbf{k}$, Stokes' theorem gives $\oint_C \mathbf{F} \cdot d\mathbf{r} = \iint_D \text{curl}(\mathbf{F}) \cdot \mathbf{k} \, dx \, dy$, and since $\text{curl}(\mathbf{F}) \cdot \mathbf{k} = \partial Q / \partial x - \partial P / \partial y$, this is exactly Green's theorem.

Chapter 4: Gauss Divergence Theorem

4.1 Statement (Translation Between Volume and Surface Integrals)

Let T be a closed bounded region in space whose boundary is a piecewise smooth surface S , and let $\mathbf{F}(x, y, z)$ be a vector function with continuous first partial derivatives in a domain containing T . Then:

$$\iiint_T \text{div } \mathbf{F} \, dV = \iint_S \mathbf{F} \cdot \mathbf{n} \, dA$$

where $\mathbf{n}$ is the outer unit normal vector to S . Writing $\mathbf{F} = F_1 \mathbf{i} + F_2 \mathbf{j} + F_3 \mathbf{k}$ and $\mathbf{n} = (\cos\alpha, \cos\beta, \cos\gamma)$, this can also be written in component form:

$$\iiint_T (\partial F_1 / \partial x + \partial F_2 / \partial y + \partial F_3 / \partial z) dx dy dz = \iint_S (F_1 \cos\alpha + F_2 \cos\beta + F_3 \cos\gamma) dA$$

or, expressed using projected area elements $dy dz$, $dz dx$, $dx dy$:

$$\iiint_T (\partial F_1 / \partial x + \partial F_2 / \partial y + \partial F_3 / \partial z) dx dy dz = \iint_S (F_1 dy dz + F_2 dz dx + F_3 dx dy)$$

4.2 Physical Significance

- The divergence of a vector field at a point may be positive, negative, or zero.
- If $\text{div } \mathbf{F} > 0$ at a point, the flux leaving the surface exceeds the flux entering — the point acts as a source.
- If $\text{div } \mathbf{F} < 0$ at a point, the flux entering exceeds the flux leaving — the point acts as a sink.
- If $\text{div } \mathbf{F} = 0$ at a point, the flux entering equals the flux leaving — the field is source-free (solenoidal) there.

Worked Example — Flux Out of a Cylinder

Evaluate $\iint_S (x^3 dy dz + x^2 y dz dx + x^2 z dx dy)$ where S is the closed surface consisting of the cylinder $x^2 + y^2 = a^2$, $0 \leq z \leq b$, together with the circular disks $z = 0$ and $z = b$.

Solution:

Here $F_1 = x^3$, $F_2 = x^2 y$, $F_3 = x^2 z$, so $\text{div } \mathbf{F} = 3x^2 + x^2 + x^2 = 5x^2$. Introducing polar coordinates $x = r \cos\theta$, $y = r \sin\theta$, $dx dy dz = r dr d\theta dz$, and applying the Divergence theorem:

$$\iiint_T 5x^2 dx dy dz = 5 \int_{[z=0 \rightarrow b]} \int_{[\theta=0 \rightarrow 2\pi]} \int_{[r=0 \rightarrow a]} r^2 \cos^2\theta \cdot r dr d\theta dz = 5b \cdot \pi \cdot (a^4/4) = 5\pi a^4 b/4$$

4.3 Practice Problems

Evaluate $\iint_S \mathbf{F} \cdot \mathbf{n} dA$ using the Divergence theorem in each case:

- $\mathbf{F} = [e^x, e^y, e^z]$, S the surface of the cube $-1 \leq x, y, z \leq 1$
- $\mathbf{F} = [x^2, y^2, z^2]$, S the sphere $x^2 + y^2 + z^2 = 9$
- $\mathbf{F} = [x, y, z]$, S the hemisphere $x^2 + y^2 + z^2 = 4$, $z \geq 0$, together with its base disk

Chapter 5: Stokes' Theorem

5.1 Statement

Let S be a piecewise smooth oriented surface with unit normal $\mathbf{n}$, bounded by a piecewise smooth simple closed curve $C = \partial S$, positively oriented relative to $\mathbf{n}$. If $\mathbf{F}$ is a continuously differentiable vector field on a domain containing S , then Stokes' theorem states:

$$\oint_C \mathbf{F} \cdot d\mathbf{r} = \iint_S (\text{curl } \mathbf{F}) \cdot \mathbf{n} dA$$

Stokes' theorem relates a line integral around C to a surface integral over any surface S having C as its boundary. Depending on convenience, either integral may be computed in terms of the other.

5.2 Applications of Stokes' Theorem

Example 1 — Computing a Line Integral via a Surface Integral

Compute $\oint_C \mathbf{F} \cdot d\mathbf{r}$, where $\mathbf{F} = y \mathbf{i} + xz^3 \mathbf{j} - zy^3 \mathbf{k}$ and C is the circle in the plane $z = -3$, $x^2 + y^2 = 4$, oriented anti-clockwise.

Solution:

To apply Stokes' theorem, choose the convenient surface S : $x^2 + y^2 \leq 4$, $z = -3$, whose boundary is C — the flat circular disc. Taking the normal $\mathbf{n} = \mathbf{k}$ gives S the anti-clockwise orientation matching C . Since:

$$\text{curl}(\mathbf{F}) \cdot \mathbf{k} = \partial/\partial x(xz^3) - \partial/\partial y(y) = z^3 - 1$$

and $z = -3$ is constant on S , $\text{curl}(\mathbf{F}) \cdot \mathbf{k} = (-3)^3 - 1 = -28$ throughout S . Therefore:

$$\oint_C \mathbf{F} \cdot d\mathbf{r} = \iint_S (\text{curl } \mathbf{F}) \cdot \mathbf{n} \, dA = \iint_S (-28) \, dA = -28 \cdot (4\pi) = -112\pi$$

Example 2 — When Stokes' Theorem Does Not Apply Directly

Consider $\mathbf{F}(x, y) = -y/(x^2+y^2) \mathbf{i} + x/(x^2+y^2) \mathbf{j}$, and let C be the unit circle $x^2 + y^2 = 1$, $z = 0$, oriented anti-clockwise. A direct computation gives $\text{curl } \mathbf{F} = \mathbf{0}$ wherever $\mathbf{F}$ is defined, and a direct evaluation of the line integral gives $\oint_C \mathbf{F} \cdot d\mathbf{r} = 2\pi$.

Note: This does not contradict Stokes' theorem. $\mathbf{F}$ is undefined at the origin $(0, 0)$, which lies inside every disc bounded by C , so no surface S contained in the domain of $\mathbf{F}$ has C as its boundary — the hypotheses of Stokes' theorem simply are not met. This is the same reason $\mathbf{F}$ fails to be conservative even though $\text{curl } \mathbf{F} = \mathbf{0}$: the domain of $\mathbf{F}$ is not simply connected.

5.3 Conditions for Conservativeness

Let D be a simply connected domain in $\mathbb{R}^3$ and $\mathbf{F}: D \rightarrow \mathbb{R}^3$ a continuously differentiable vector field. The following statements are equivalent:

- $\mathbf{F}$ is conservative ($\mathbf{F} = \text{grad } \phi$ for some scalar potential ϕ).
- $\int_C \mathbf{F} \cdot d\mathbf{r}$ is independent of the path $C(A, B)$ joining A to B .
- $\text{curl } \mathbf{F} = \mathbf{0}$ throughout D .

The implication $\text{curl } \mathbf{F} = \mathbf{0} \Rightarrow \text{path-independence}$ follows from Stokes' theorem: for any simple closed curve C in D , simple connectedness guarantees an oriented piecewise smooth surface $S \subset D$ with $\partial S = C$, so $\oint_C \mathbf{F} \cdot d\mathbf{r} = \iint_S \text{curl}(\mathbf{F}) \cdot \mathbf{n} \, dA = 0$.

5.4 Physical Interpretation of Curl

Stokes' theorem gives a way of interpreting $\text{curl } \mathbf{F}$ in the context of fluid flow. Consider a small circular disc S_a of radius a , centred at a point P_0 in the domain of $\mathbf{F}$, with unit normal $\mathbf{n}$. By Stokes' theorem:

$$\oint_{S_a} (\mathbf{F} \cdot \mathbf{T}) \, ds = \iint_{S_a} (\text{curl } \mathbf{F}) \cdot \mathbf{n} \, dA \approx (\text{curl } \mathbf{F}(P_0) \cdot \mathbf{n}) \cdot A(S_a)$$

The quantity $\mathbf{F} \cdot \mathbf{T}$ is the component of $\mathbf{F}$ along the tangent to the boundary circle, so $\oint (\mathbf{F} \cdot \mathbf{T}) \, ds$ is the circulation of $\mathbf{F}$ around C_a . Letting $a \rightarrow 0$:

$$(\text{curl } \mathbf{F})(P_0) \cdot \mathbf{n} = \lim_{a \rightarrow 0} \{ (1/A(S_a)) \oint_{S_a} (\mathbf{F} \cdot \mathbf{T}) \, ds \}$$

For this reason, the normal component of $\text{curl } \mathbf{F}$ is called the specific circulation of the fluid at P_0 , and is greatest when $\mathbf{n}$ is aligned with $\text{curl } \mathbf{F}(P_0)$. If $\text{curl } \mathbf{F} = \mathbf{0}$ throughout a region, there is no circulation tendency anywhere in it, and the fluid is said to be irrotational.

5.5 Practice Exercises

Using Stokes' theorem, evaluate $\oint_C \mathbf{F} \cdot d\mathbf{r}$ in each case:

- $\mathbf{F}(x, y, z) = 2y \mathbf{i} + 3z \mathbf{j} + x \mathbf{k}$, and C is the triangle with vertices $A(0,0,0)$, $B(0,2,0)$, $C(1,1,1)$, oriented $A \rightarrow B \rightarrow C \rightarrow A$
- $\mathbf{F}(x, y, z) = z^2 \mathbf{i} + y \mathbf{j} + xz \mathbf{k}$, and C is the boundary of the upper hemisphere $S: z = \sqrt{4 - x^2 - y^2}$, oriented counter-clockwise
- $\mathbf{F}(x, y, z) = x^2 \mathbf{i} + 4xy^2 \mathbf{j} + y^2x \mathbf{k}$, and C is the rectangle through $A(0,0,0)$, $B(1,0,0)$, $C(1,3,2)$, $D(0,3,2)$, oriented $A \rightarrow B \rightarrow C \rightarrow D \rightarrow A$
- $\mathbf{F}(x, y, z) = 3y \mathbf{i} + 4x \mathbf{j} + 2y \mathbf{k}$, and C is the boundary of the paraboloid $z = 4 - x^2 - y^2$, $z \geq 0$, oriented counter-clockwise

Quick Reference — All Key Formulae (Module 4)

Surface Integrals

$$\iint_S \mathbf{F} \cdot \mathbf{n} \, dA; \text{ reduces to } \iint_R \mathbf{F} \cdot \mathbf{n} \sqrt{1+f_x^2+f_y^2} \, dx \, dy \text{ for } z = f(x,y)$$

Volume Integrals

$$\iiint_T f \, dV; \, dV = r \, dr \, d\theta \, dz \text{ (cylindrical)}, \, dV = r^2 \sin\phi \, dr \, d\theta \, d\phi \text{ (spherical)}$$

Green's Theorem

$$\oint_C (P \, dx + Q \, dy) = \iint_R (\partial Q / \partial x - \partial P / \partial y) \, dx \, dy; \text{ Area} = (1/2) \oint_C (x \, dy - y \, dx)$$

Gauss Divergence Theorem

$$\iiint_T \text{div } \mathbf{F} \, dV = \iint_S \mathbf{F} \cdot \mathbf{n} \, dA$$

Stokes' Theorem

$$\oint_C \mathbf{F} \cdot d\mathbf{r} = \iint_S (\text{curl } \mathbf{F}) \cdot \mathbf{n} \, dA$$

Conservativeness (simply connected D)

$$\mathbf{F} \text{ conservative} \Leftrightarrow \text{path-independent} \Leftrightarrow \text{curl } \mathbf{F} = 0$$

Curl as Circulation Density

$$(\text{curl } \mathbf{F})(P_0) \cdot \mathbf{n} = \lim_{a \rightarrow 0} \{ (1/A(S_a)) \oint_{C_a} (\mathbf{F} \cdot T) \, ds$$

— End of Module 4 Notes —